

Chapter-1-

Error in Measurement



TYPES OF ERRORS

Different types of errors that occur in measurements have been classified into three categories. These include

- Gross errors.
- Systematic errors.
- Random errors.



Gross Errors

These are basically human errors caused by the operator or person using the instrument. The instrument may be good and may not give any error but still the measurement may go wrong due to the operator. The different types of gross errors are:

- Taking wrong readings.
- Reading with parallax error.
- Incorrect adjustments of zero and full-scale adjustments.
- Improper applications of instruments: Using a 0–100 V voltmeter to measure 0.1 V, etc.
- Wrong computation: When power is to be determined, ' V ' and ' I ' are measured. If the computation goes wrong, the measurement of power will be wrong. Thus, wrong computation can result in error.

Systematic Errors

These are divided into two categories:

Instrumental errors: Due to shortcomings of the instruments.

Environmental errors: Due to external conditions affecting the instrument.

Instrumental errors.

Even if human errors are avoided or proper care is taken to see that such errors do not occur, errors can still occur in measurements due to the instrument.

The possible reasons can be as follows:



- a. Friction in bearings of various moving components can cause incorrect readings.
- b. Irregular spring tension in analog meters.
- c. Calibration errors due to aging.
- d. Zero setting not adjusted properly.
- e. Full-scale setting not adjusted properly.
- f. Faulty display circuit in digital instruments.

These errors can be avoided by

- Applying correction factors.
- Selecting suitable instruments.
- Calibrating the instruments.



Environmental errors.

Ambient parameters such as temperature, pressure, humidity, magnetic and electrostatic fields, dust, and other such external parameters can affect the performance of the instrument. Improper housing of the instrument also can give wrong readings. Such errors can be avoided by air-conditioning, magnetic shielding, cleaning the instruments, and housing the instruments properly depending on the application and type of the instrument.



Random Errors

Though gross errors and systematic errors can be avoided by taking proper care, some other errors can also occur in measurements. No specific reason can be assigned and precaution could be taken to avoid these errors.

Such errors are categorized as random errors. Noise that is impracticable can cause random errors in measurements. To avoid these errors, frequency of measurement is to be increased, i.e., the same parameter is to be measured often. Error in such measurements can be estimated by statistical analysis.



Absolute Error (e_{ab})

It is defined as the difference between the expected value of the variable and the measured value of the variable

$$e = Y_n - X_n$$

Where:

e = absolute error

Y_n = expected value

X_n = measured value



If we wish to express the error as a percentage

$$\text{percent error} = \frac{\text{Absolute error}}{\text{Expected value}} (100)$$

$$\text{percent error} = \frac{e}{Y_n} (100)$$

$$\text{percent error} = \left| \frac{Y_n - X_n}{Y_n} \right| (100)$$



Relative Accuracy (A)

Is defined as the ratio between the measured value and the expected value

$$A = \frac{X_n}{Y_n} = 1 - \left| \frac{Y_n - X_n}{Y_n} \right|$$

Accuracy expressed as percent accuracy, a, is

$$a = 100\% - \text{percent error} = A \times 100$$



Example

The expected value of the voltage across a resistor is 50V., however, measurement yields a value of 49V. Calculate

- (a) The absolute error
- (b) The percent of error
- (c) The relative accuracy
- (d) The percent of accuracy

Solution

(a)

$$e = Y_n - X_n = 50V - 49V = 1V$$

(b)

$$\text{percent error} = \frac{50V - 49V}{50V} \times 100\% = \frac{1}{50} \times 100\% = 2\%$$

(c)

$$A = 1 - \frac{50V - 49V}{50V} = 1 - \frac{1}{50} = 0.98$$

(d)

$$a = 100\% - \text{percent error} = 100\% - 2\% = 98\%$$

$$= A \times 100 = 0.98 \times 100 = 98\%$$



Precision

The Precision of a measurement is a numerical indication of the closeness with which a repeated set of measurements of the same variable agrees with the average of the set of measurements

$$\text{Precision} = 1 - \left| \frac{X_n - \overline{X_n}}{\overline{X_n}} \right|$$

Where:

X_n = the value of the nth measurement

$\overline{X_n}$ = the average of the set of n measurements



Example

The following set of ten measurements was recorded in the laboratory. Calculate the precision of the fourth measurement.

Measurement Number	Measurement Value X_n (volts)
1	98
2	102
3	101
4	97
5	100
6	103
7	98
8	106
9	107
10	99



Solution

$$\text{The average value} = \frac{\text{Sum of the ten measurements}}{10} = 101.1$$

$$\text{Precision} = 1 - \left| \frac{X_n - \overline{X_n}}{\overline{X_n}} \right|$$

$$\text{Precision} = 1 - \left| \frac{97 - 101.1}{101.1} \right| = 1 - 0.04 = 0.96$$



Statistical Analysis of Errors in Measurements

- When we measure any physical quantity, our measurements are affected by a multitude of factors.
- If a measurement is repeated several times, the readings may differ because of the experimenter's inability to keep all the above factors constant.
- Statistical Analysis allow us to obtain such information as the mean value, average deviation , and standard deviation of our data. Such information allows us to make quantitative judgments of the variations, or errors in our data.



Average value (arithmetic mean)

The arithmetic mean of a set of n numbers $x_1, x_2, \dots, \bar{x}_n$ is denoted by $\bar{x}$ and is defined as

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

Deviation

Is the difference between each piece of test data and the arithmetic mean. The deviation of $x_1, x_2, \dots$ from their arithmetic mean $\bar{x}$ is denoted by $d_1, d_2, \dots, d_n$ and is defined as

$$d_1 = x_1 - \bar{x}$$

$$d_2 = x_2 - \bar{x}$$

$$d_n = x_n - \bar{x}$$

The average deviation D

$$D = \frac{|d_1| + |d_2| + |d_3| + \dots + |d_n|}{n}$$

The standard deviation S

The standard deviation S for a set of values is the degree to which the values vary about the average value

$$S = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2}{n \text{ or } (n - 1)}}$$

For $n < 30$ use $(n-1)$

For $n \geq 30$ use n



Example

For the following data compute.

- (a) the arithmetic mean.
- (b) The deviation of each value
- (c) The algebraic sum of the deviations.
- (d) The average deviation
- (e) The standard deviation

variable	value
X_1	50.1
X_2	49.7
X_3	49.6
X_4	50.2

Solution

- (a) the arithmetic mean is computed as

$$\bar{x} = \frac{50.1 + 49.7 + 49.6 + 50.2}{4} = \frac{199.6}{4} = 49.9$$



(b) The deviation of each value from the mean is computed as

$$d_1 = 50.1 - 49.9 = 0.2$$

$$d_2 = 49.7 - 49.9 = -0.2$$

$$d_3 = 49.6 - 49.9 = -0.3$$

$$d_4 = 50.2 - 49.9 = 0.3$$

(c) The algebraic sum of the deviations is

$$d_{tot} = 0.2 - 0.2 + 0.3 - 0.3 = 0$$

(d) The average deviation is computed as

$$D = \frac{|0.2| + |-0.2| + |-0.3| + |0.3|}{4} = \frac{1.0}{4} = 0.25$$

(e) The standard deviation is computed as

$$S = \sqrt{\frac{(0.2)^2 + (-0.2)^2 + (-0.3)^2 + (0.3)^2}{(4 - 1)}} = \sqrt{\frac{0.26}{3}} = 0.294$$

Limiting Errors

Most manufactures of measuring instruments state that an instrument is accurate within a certain percentage of full-scale reading

Example

A 300V voltmeter is specified to be accurate within $\pm 2\%$ at full scale. Calculate the limiting error when the instrument is used to measure a 120V source.

Solution

the limiting error is

$$0.02 \times 300V = 6V$$

Therefore, the limiting error at 120V is

$$\frac{6}{120} \times 100\% = 5\%$$

